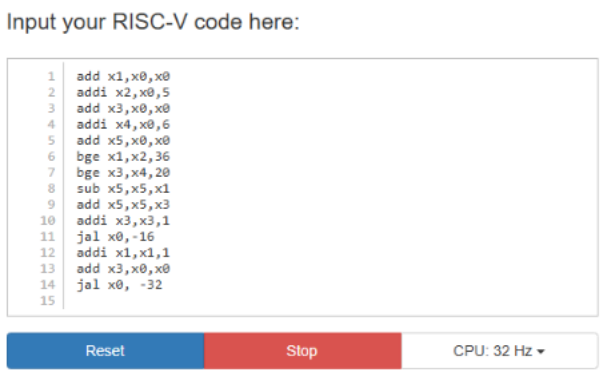
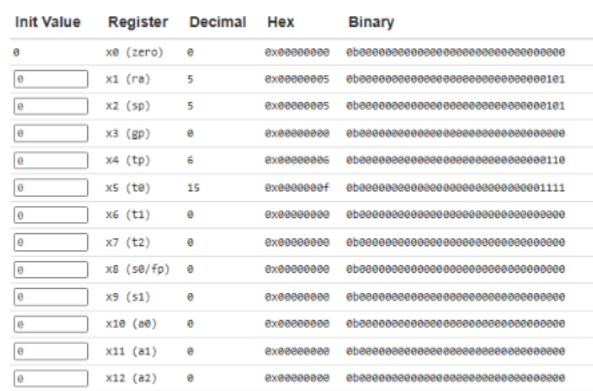
Test Cases:

- x1 = 0, x2 = 5, x3 = 0, x4 = 6, x5 = 0 --> x1 = 5, x3 = 0, x5 = 15

For the outer loop, x1 is compared with x2 using x1 >= x2. With the x2 register having the value of 5, x1 = 0 meaning it isn't greater than nor equal to 5. From the outer loop being false, it would then go to the inner loop where x3 is being compared to x4 using x3 >= x4. With x4 register having the value of 6, x3 = 0 meaning it isn't greater than nor equal to 6. With the inner loop false, it wouldn't skip any steps and start the actions within the loop. X5 = 0, subtracts as x5 = x5 – x1 = 0-0 = 0, but then executes addition with x5 = x5 + x3 = 0+0 with x5, s, value being 0 for now. X3 value would then be incremented by one, then goes to jal x0, -16 taking the updated x3 value to be compared to x4 as x3 >= x4 again. Once x3 updates it being equal to x4 = 6, it would skip the actions in the inner loop. X1 would then be incremented by 1 then x3 resets back to 0 because once jal x0, -32 executes it jumps back to the outer loop checking x1 >= x2 and then the inner loop with x3>=x4. When x1 = 4 and x3 = 5, it would be the last action within the inner loop as S = 15 (S = 14-4+5 = 15). Once x1 = 5, which is the maximum for the outer loop, the program will be finished.





i	j	S
0	0	S = 0-0+0 = 0
0	1	S = 0-0+1 = 1
0	2	S = 1-0+2 = 3
0	3	S = 3-0+3 = 6
0	4	S = 6-0+4 = 10
0	5	S = 10-0+5 = 15
1	0	S = 15-1+0 = 14
1	1	S = 14-1+1 = 14
1	2	S = 14-1+2 = 15
1	3	S = 15-1+3 = 17
1	4	S = 17-1+4 = 20
1	5	S = 20-1+5 = 24
2	0	S = 24-2+0 = 22

2	1	S = 22-2+1 = 21
2	2	S = 21-2+2 = 21
2	3	S = 21-2+3 = 22
2	4	S = 22-2+4 = 24
2	5	S = 24-2+5 = 27
3	0	S = 27-3+0 = 24
3	1	S = 24-3+1 = 22
3	2	S = 22-3+2 = 21
3	3	S = 21-3+3 = 21
3	4	S = 21-3+4 = 22
3	5	S = 22-3+5 = 24
4	0	S = 24-4+0 = 20
4	1	S = 20-4+1 = 17
4	2	S = 17-4+2 = 15
4	3	S = 15-4+3 = 14
4	4	S = 14-4+4 = 14
4	5	S = 14-4+5 = 15

- x1 = 0, x2 = 10, x3 = 0, x4 = 5, x5 = 0 --> x1 = 10, x3 = 0, x5 = -125

For the outer loop, x1 is compared with x2 using x1 >= x2. With the x2 register having the value of 10, x1 = 0 meaning it isn't greater than nor equal to 10. From the outer loop being false, it would then go to the inner loop where x3 is being compared to x4 using x3 >= x4. With x4 register having the value of 5, x3 = 0 meaning it isn't greater than nor equal to 5. With the inner loop false, it wouldn't skip any steps and start the actions within the loop. X5 = 0, subtracts as x5 = x5 – x1 = 0-0 = 0, but then executes addition with x5 = x5 + x3 = 0+0 with x5, s, value being 0 for now. X3 value would then be incremented by one, then goes to jal x0, -16 taking the updated x3 value to be compared to x4 as x3 >= x4 again. Once x3 updates to being equal to x4 = 5, it would skip the actions in the inner loop. X1 would then be incremented by 1 then x3 resets back to 0 because once jal x0, -32 executes it jumps back to the outer loop checking x1 >= x2 and then the inner loop with x3>=x4. When x1 = 9 and x3 = 4, it would be the last action within the inner loop as S = -125 (S = -120-9+4 = -125). Once x1 = 10, which is the maximum for the outer loop, the program will be finished.

Input your RISC-V code here:

```
add x1,x0,x0
addi x2,x0,10
add x3,x0,x0
addi x4,x0,5
add x5,x0,x0
bge x1,x2,36
bge x3,x4,20
sub x5,x5,x1
add x5,x5,x3
addi x3,x3,1
jal x0,-16
addi x1,x1,1
add x3,x0,x0
jal x0, -32
```

Reset | Stop | CPU: 32 Hz

Init Value	Register	Decimal	Hex	Binary
0	x0 (zero)	0	0x00000000	0b00000000000000000000000000000000
0	x1 (ra)	10	0x0000000a	0b00000000000000000000000000001010
0	x2 (sp)	10	0x0000000a	0b00000000000000000000000000001010
0	x3 (gp)	0	0x00000000	0b00000000000000000000000000000000
0	x4 (tp)	5	0x00000005	0b00000000000000000000000000000101
0	x5 (t0)	-125	0xffffff83	0b11111111111111111111111110000011
0	x6 (t1)	0	0x00000000	0b00000000000000000000000000000000
0	x7 (t2)	0	0x00000000	0b00000000000000000000000000000000
0	x8 (s0/fp)	0	0x00000000	0b00000000000000000000000000000000
0	x9 (s1)	0	0x00000000	0b00000000000000000000000000000000
0	x10 (a0)	0	0x00000000	0b00000000000000000000000000000000
0	x11 (a1)	0	0x00000000	0b00000000000000000000000000000000
0	x12 (a2)	0	0x00000000	0b00000000000000000000000000000000

i	j	s
0	0	S = 0-0+0 = 0
0	1	S = 0-0+1 = 1
0	2	S = 1-0+2 = 3
0	3	S = 3-0+3= 6
0	4	S = 6-0+4 = 10
1	0	S = 10-1+0 = 9
1	1	S = 9-1+1 = 9
1	2	S = 9-1+2 = 10
1	3	S = 9-1+3 = 12
1	4	S = 9-1+4 = 15
2	0	S = 15-2+0 = 13
2	1	S = 13-2+1 = 12
2	2	S = 12-2+2 = 12
2	3	S = 12-2+3 = 13
2	4	S = 13-2+4 =15
3	0	S = 15-3+0 = 12
3	1	S = 12-3+1 = 10
3	2	S = 10-3+2 = 9
3	3	S = 9-3+3 = 9
3	4	S = 9-3+4 = 10
4	0	S = 10-4+0 = 6
4	1	S = 6-4+1 = 3
4	2	S = 3-4+2 = 1
4	3	S = 1-4+3 = 0
4	4	S = 0-4+4 = 0
5	0	S = 0-5+0 = -5
5	1	S = -5-5+1 = -9
5	2	S = -9-5+2 = -12
5	3	S = -12-5+3 = -14
5	4	S = -14-5+3 = -15

6	0	S = -15-6+0 = -21
6	1	S = -21-6+1 = -26
6	2	S = -26-6+2 = -30
6	3	S = -30-6+3 = -33
6	4	S = -33-6+4 = -35
7	0	S = -35-7+0 = -42
7	1	S = -42-7+1 = -48
7	2	S = -48-7+2 = -53
7	3	S = -53-7+3 = -57
7	4	S = -57-7+4 = -60
8	0	S = -60-8+0 = -68
8	1	S = -68-8+1 = -75
8	2	S = -75-8+2 = -81
8	3	S = -81-8+3 = -86
8	4	S = -86-8+4 = -90
9	0	S = -90-9+0 = -99
9	1	S = -99-9+1 = -107
9	2	S = -107-9+2 = -114
9	3	S = -114-9+3 = -120
9	4	S = -120-9+4 = -125